RAJALAKSHMI ENGINEERING COLLEGE

Kirankumar V 250701354-2025-CSE

K2

Started on	Wednesday, 16 September 2026, 9:18 AM
State	Finished
Completed on	Wednesday, 16 September 2026, 9:27 AM
Time taken	9 mins 3 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array of N integer, we have to maximize the sum of arr[i] * i, where i is the index of the element (i = 0, 1, 2, ..., N).Write an algorithm based on Greedy technique with a Complexity O(nlogn).

Input Format:

First line specifies the number of elements-n

The next n lines contain the array elements.

Output Format:

Maximum Array Sum to be printed.

Sample Input:

5

2 5 3 4 0

Sample output:

40

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int ar[n];
    for(int i=0;i<n;i++)
        scanf("%d",&ar[i]);
    for(int i=0;i<n;i++){
        for(int j=i+1;j<n;j++){
            if(ar[i]>=ar[j]){
                int t = ar[i];
                ar[i] =ar[j];
                ar[j]=t;
            }
        }
    }
    /*for(int i=0;i<n;i++)
        printf("%d ",ar[i]);*/
    int s=0;
    for(int i=0;i<n;i++){
        s=s+(ar[i]*i);
    }
    printf("%d",s);

}
```

	Input	Expected	Got	
✔	5 2 5 3 4 0	40	40	✔

	Input	Expected	Got	
✔	10 2 2 2 4 4 3 3 5 5 5	191	191	✔
✔	2 45 3	45	45	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course